msinv: ARG-based coalescent simulator with chromosomal inversions

Feature	msinv	msprime	discoal	SLiM
Chromosomal inversions (any ag	✓	✗	✗	✓
Cross-karyotype barrier (t_inv)	✓	✗	✗	✓
Gene flux (Peischl phi(x))	✓	✗	✗	partial
Multiple inversions (incl. nested	✓	✗	✗	✓
Per-pop inversion frequencies	✓	✗	✗	✓
Selective sweeps	✓	partial	✓	✓
Coalescent (fast, neutral)	✓	✓	✓	✗
ms-style demography	✓	✓	✓	partial
Multiple populations	✓	✓	✓	✓
Tree sequence (tskit) output	✓	✓	✗	✓

Figure 8. Feature comparison: msinv vs existing coalescent and forward simulators. msinv is the only coalescent simulator with explicit chromosomal inversion support — cross-karyotype barriers (t_inv), position-dependent gene flux ($\phi(x)$), multiple/nested inversions, and per-population frequencies. Unlike SLiM (forward-time), msinv is coalescent-based: fast for neutral scenarios and directly produces tree sequences for downstream analysis with tskit. msinv v0.3.0, hull algorithm (per-position ancestral material tracking).
Command: pixi run -e all python examples/make_figures.py